MIL-T-14120A
28 January 1960
SUPERSEDING
MIL-P-14120(Ord)
18 July 1955

MILITARY SPECIFICATION

THERMOMETERS, POWDER TEMPERATURE, M1 AND M1A1; PACKAGING OF

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the packaging and packing of Powder Temperature Thermometers M1 and M1A1, with or without a carrying case, for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive, Adhesive, Paper, Water Resistant

FSC 6685

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

3.1.1 Materials.- The following shall apply wherever packaging materials are referred to in this specification by name only:

REFERENCE	FULL IDENTIFICATION
Gummed tape	Class 2 of Specification UU-T-111
Water-vaporproof bag	Type II, class e of Specification MIL-B-117

3.2 Size of containers.- Bags, interior containers and exterior containers shall be of a minimum size that will adequately contain the items.

3.3 Packaging.

3.3.1 Level A.

3.3.1.1 Cleaning.- Cleaning process C-1 of Specification MIL-P-116 shall be followed to clean the thermometer and the carrying case.

3.3.1.2 Drying.- Immediately after cleaning, items shall be thoroughly dried in accordance with any applicable procedure, specified in Specification MIL-P-116

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

that will not damage the items.

3.3.1.3 Unit packaging.- Packaging shall be as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

3.3.1.3.1 Packaging the thermometer without a carrying case.

3.3.1.3.1.1 Covering glass surface.- The glass surface of the thermometer dial shall be covered with at least three thicknesses of lens tissue conforming to Specification MIL-P-13988. The tissue shall be secured in place with pressure-sensitive tape conforming to type I or II of Specification UU-T-106.

3.3.1.3.1.2 Immobilizing the thermometer in a set-up box.- The thermometer shall be immobilized in a set-up paperboard box by utilizing three pieces of chipboard as shown in Figure 1. The chipboard shall be at least .033 inch thick and conforming to the chipboard specified in Specification PPP-B-676. The box shall conform to type I or IV, any class and style of Specification PPP-B-676. The inside dimensions of the box shall not be more than 7 1/2 inches by 2. inches by 2. inches. The box shall be closed by covering the seams with gummed tape.

3.3.1.3.1.3 Water-vaporproof bag.- The boxed thermometer shall be placed in a water-vaporproof bag. The corners of the box shall be blunted prior to placing the box in the bag. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

3.3.1.3.2 Packaging the thermometer with a carrying case.- The thermometer shall be placed in its carrying case and the cover positioned and secured in place. The carrying case shall then be wrapped with paperboard cushioning conforming to type I, class 1 of Specification PPP-P-291. The wrap shall be secured in place with gummed tape. The wrapped carrying case shall then be placed in a water-vaporproof bag. The bag shall be heat sealed and the air exhausted as specified in 3.3.1.3.1.3.

3.3.1.4 Intermediate packaging.- Unless otherwise specified, only items with the same stock number shall be placed in an intermediate container. Twenty unit packaged thermometers, with or without carrying cases, shall be immobilized in a fiberboard box conforming to class 2 or 3, any compliance symbol and any style of Specification PPP-B-636. The fiberboard box shall be closed with pressure-sensitive tape conforming to Specification PPP-T-76 in accordance with the appendix of Specification PPP-B-636.

3.3.2 Level C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

3.3.2.1 Cleaning and drying.- Cleaning and drying shall be as specified in 3.3.1.1 and 3.3.1.2.

3.3.2.2 Unit packaging.- The packaging procedures for thermometers, with or without carrying cases, shall be such as to afford adequate protection against corrosion and physical damage during shipment and interim storage prior to use at the first receiving installation.

3.4 Packing.- Unless otherwise specified, all the items which are packed in a shipping container shall contain the same stock number.

3.4.1 Level A.- Twenty-four fiberboard boxes containing thermometers without carrying cases or six fiberboard boxes containing thermometers with carrying cases shall be immobilized in a shipping container conforming to either one of the following specifications:

PPP-B-601 - Style A or B (overseas type) - type II or III, class 2 plywood
PPP-B-621 - Class 2 (overseas), style 2 or 4

3.4.2 Level B.- Twenty-four fiberboard boxes containing thermometers without carrying cases or six fiberboard boxes containing thermometers with carrying cases shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), style 2 or 4

3.4.3 Level C.- A number of packaged thermometers, with or without carrying cases, shall be immobilized in a shipping container conforming to type I or II, class 1, any style, except FTC, TS, OPF and FPF of Specification PPP-B-636. The gross weight for a fiberboard shipping container shall not exceed the limits of the fiberboard box specification.

3.4.4 Strapping.- Strapping shall be applied on wood and plywood shipping containers. The type of strapping and the method of applying it shall be as specified in the applicable container specification.

3.5 Marking.

3.5.1 Interior packages.- Each bag and each box shall be marked in accordance with Standard MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

3.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. Fragile markings shall be applied as specified in Standard MIL-STD-129.

3.5.3 Stock number.- The stock number to be added in the identification data shall be obtained from the contracting officer.

3.6 Pilot pack.- Prior to quantity production, a pilot pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification for Level A. A rough handling test is required.

3.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the specification and other documents referenced in this specification. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that the substitute procedures will provide, as a minimum, the quality assurance required in the contractual documents. In cases of dispute as to whether certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced therein.

Source: https://assist.dla.mil - Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

4.1.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection, when deemed necessary by the Government, to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an affect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.1.4 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.1.5 Pilot lot (assemblied item).- If a pilot lot is specified in the invitation for bid or by the contracting officer, it shall be subjected to 100 percent inspection by the contractor or his agent and verified by the Government inspector to determine acceptability in accordance with the quality assurance provisions outlined herein.

4.1.6 Pilot lot failure.- Should the pilot lot fail to pass the Government acceptance test, production of those components causing rejection shall cease until the defective characteristics have been corrected in the process used to produce the pilot lot. When corrections have been made, a new lot, the size of which shall be determined by the contracting officer, shall be completely inspected by the contractor and witnessed by the Government inspector. All rejected lots shall be corrected by the contractor and resubmitted as parts of the first production lot submitted for Government acceptance.

4.2 Inspection provisions.- Quality assurance provisions as specified in Specification MIL-P-116 shall be utilized, where applicable.

4.2.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.2.2 Examination.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

4.2.2.1 All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS
(Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105)

		Req. Par.	Test Method
Major	AQL 2.5		
1.	Marking illegible or missing (Levels A, B & C)	3.5	Visual
2.	Improper cleaning (Levels A & C)	3.3.1.1 3.3.2.1	Visual and MIL-P-116
3.	Improper drying (Levels A & C)	3.3.1.2 3.3.2.1	Visual
4.	Item not immobilized (Levels A & C)	3.3.1.3.1.2 3.3.1.4 3.4.1	Visual
5.	Water-vaporproof bag damaged, improperly sealed (Level A)	3.3.1.3.1.3 3.3.1.3.2	Visual
Minor	AQL 4.0		
41.	Tissue missing (Level A)	3.3.1.3.1.1	Visual
42.	Fiberboard wrapping omitted, not secured (Level A)	3.3.1.3.2	Visual
43.	Tape omitted, not properly applied (Level A)	3.3.1.3.1.2 3.3.1.3.2 3.3.1.4	Visual
44.	Stock numbers different in same container (Level A)	3.3.1.4 3.4	Visual
45.	Non-conformance of container (Levels A, B & C)	3.4.1; 3.4.2 3.4.3	Visual
46.	Strapping improperly applied (Levels A, B & C)	3.4.4	Visual

TABLE II - CLASSIFICATION OF DEFECTS
(Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105)

		Method of Protection	Test Method
Major	AQL 4.0		
1.	Quick leak (Level A)	IA-8	Visual & MIL-P-116
2.	Heat seal (Level A)	IA-8	Visual & MIL-P-116

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

4.3 Test methods and procedures.

4.3.1 Rough handling test.- The pilot pack, specified in 3.6 shall be subjected to the rough handling test in accordance with the requirements of Specification MIL-P-116, applicable only to Level A.

4.3.2 Packaging tests.- The packaging tests shall be conducted in accordance with the test procedures outlined in Specification MIL-P-116, and Table II.

5. PREPARATION FOR DELIVERY

There are no applicable requirements.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing.

6.2 Size and weight of packed shipping container.

6.2.1 Thermometers without carrying case.- The following inside dimensions, cubage and gross weight are approximate for twenty-four intermediate packages packed in a nailed wood box for Level A:

Inside dimensions	- 32. inches long by 24. inches wide by 20. inches deep
Cubage	- 11. cubic feet
Gross weight	- 150. pounds

6.2.2 Thermometers with carrying case.- The following inside dimensions, cubage and gross weight are approximate for six intermediate packages packed in a nailed wood box for Level A:

Inside dimensions	- 24. inches long by 20. inches wide by 8. inches deep
Cubage	- 3.1 cubic feet
Gross weight	- 150. pounds

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.

not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

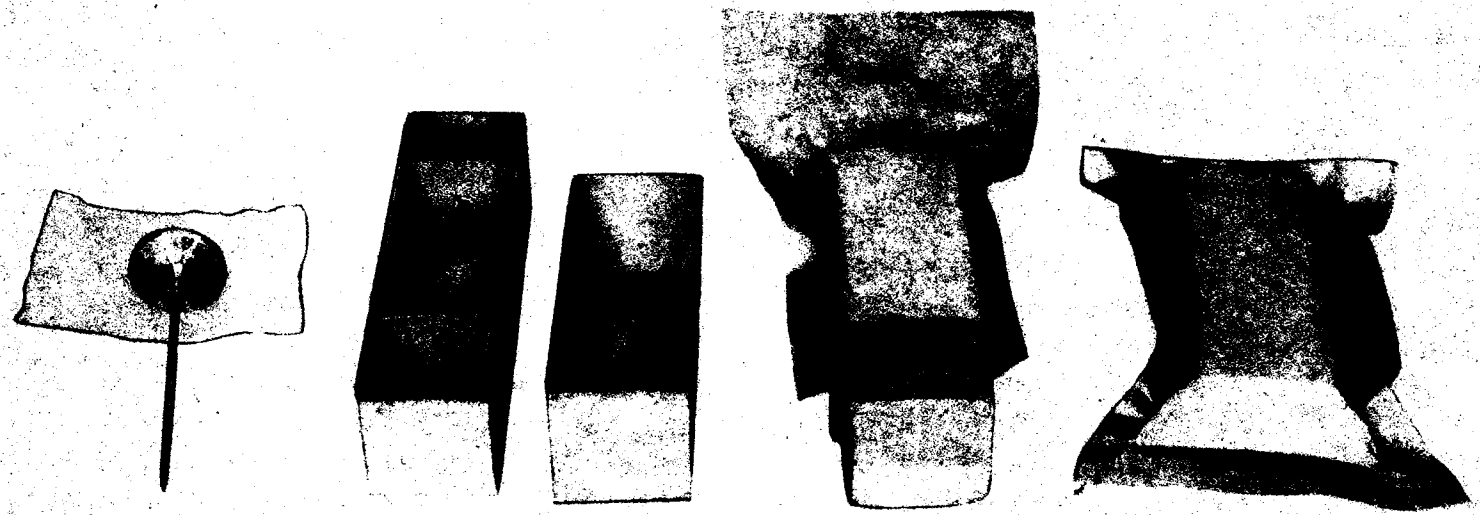
Custodians:

Army-Ordnance Corps
Navy-Marine Corps

Preparing Activity:

Army-Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:52Z
Check the source to verify that this is the current version before use.



NEG. #25615

THERMOMETER, POWDER TEMPERATURE, M1A1 - Unit packaging the thermometer without a carrying case
FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:58Z
Check the source to verify that this is the current version before use.